

ORIGINAL RESEARCH ARTICLE

Inhibition of Programmed Cell Death-1 in Cytotoxic CD8⁺ T Cells Exacerbates Pressure Overload–Induced Cardiac Injury

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BACKGROUND: The use of anti-programmed cell death-1 (PD-1) antibody increases heart failure (HF) risk in patients with cancer with preexisting cardiovascular conditions. However, the underlying mechanism remains incompletely understood.

METHODS: To evaluate the effects of anti-PD-1 antibody on transverse aortic constriction (TAC)–induced cardiac remodeling and HF, anti-PD-1 antibody–treated mice; T cell–, myeloid–, and CD8⁺ T cell–specific *Pdcd1* knockout; C-X-C motif chemokine receptor 3 (*Cxcr3*) knockout; and granzyme B (*Gzmb*) knockout mice combined with flow cytometry, Western blotting, immunofluorescence staining, pharmacological approaches, and bulk RNA-sequencing analyses were used.

RESULTS: Administration of anti-PD-1 antibody, T cell–, or CD8⁺ T cell–specific *Pdcd1* deletion, but not myeloid-specific *Pdcd1* knockout, aggravated TAC-induced cardiomyopathy and HF in mice. Mechanistically, PD-1 blockade or deletion increased myocardial infiltration of CXCR3⁺ CD8⁺ T cells, leading to granzyme B/perforin-mediated impairment of cardiomyocyte mitochondrial complex I to exacerbate TAC-induced cardiac injury and HF. TAC-enhanced chemotaxis, between cardiac fibroblast–derived CXCL9/CXCL10 and CXCR3⁺ CD8⁺ T cells, was a driving force for recruiting CXCR3⁺ CD8⁺ T cells under the conditions of PD-1 blockade or deletion. The worsened TAC-induced cardiomyopathy caused by anti-PD-1 antibody or T cell–specific *Pdcd1* deletion was rescued by genetic deletion or pharmacological blockade of granzyme B and CXCR3.

CONCLUSIONS: Anti-PD-1 antibody enhances myocardial infiltration of CXCR3⁺ CD8⁺ T cells under TAC condition through CXCL9/CXCL10-mediated chemotaxis. The increased granzyme B and perforin likely derived from CD8⁺ T cells impair function of mitochondrial respiratory chain complexes to cause cardiomyocyte apoptosis, thereby exacerbating TAC-induced cardiomyopathy and HF. CXCL9/CXCL10–CXCR3⁺ CD8⁺ T cell axis may represent a promising target for combating anti-PD-1 antibody–associated cardiotoxicity.

Key Words: cardiomyopathies ■ granzymes ■ heart failure ■ NDUFV1 protein, mouse ■ programmed cell death-1

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Clinical Perspective

What Is New?

- We uncovered the moonlighting function of programmed cell death-1 expressed in CD8⁺ T cells, but not in myeloid cells, in regulating cardiac remodeling under hemodynamic stress.
- Anti-programmed cell death-1 antibody exacerbates TAC-induced cardiac remodeling and HF exclusively through promoting myocardial infiltration of CXCR3⁺ CD8⁺ T cells.
- The CXCL9/CXCL10-CXCR3 chemotaxis-mediated myocardial infiltration of CD8⁺ T cells worsens anti-programmed cell death-1 antibody-induced cardiac injury under transverse aortic constriction, which is likely associated with the increased granzyme B and perforin.

What Are the Clinical Implications?

- Targeting the CXCL9/CXCL10-CXCR3 axis may be a novel therapeutic strategy for preventing or treating immune checkpoint inhibitor-associated cardiotoxicity.
- Cardiac-specific granzyme B inhibition may mitigate immune checkpoint inhibitor-associated cardiotoxicity.

Nonstandard Abbreviations and Acronyms

Ang II	angiotensin II
CF	cardiac fibroblast
DCM	dilated cardiomyopathy
GZMB	granzyme B
HF	heart failure
HW	heart weight
ICI	immune checkpoint inhibitor
IgG	immunoglobulin G
KO	knockout
LV	left ventricular
NDUFV1	NADH dehydrogenase (ubiquinone) flavoprotein 1
NMCM	neonatal mouse cardiomyocyte
NT-proBNP	N-terminal pro-B-type natriuretic peptide
PD-1	programmed cell death-1
RNA-seq	RNA sequencing
ROS	reactive oxygen species
TAC	transverse aortic constriction
TUNEL	TdT-mediated dUTP nick-end labeling

Hear failure (HF), the terminal stage of various heart diseases, is one of the leading causes of morbidity and mortality worldwide.^{1,2} Previous studies have shown that T lymphocytes play a critical role in the progression of HF.^{3,4} Both genetic deletion of T-cell receptor and pharmacological depletion of T cells significantly ameliorated cardiac dysfunction in mouse models of HF.⁵ Moreover, inhibition of CD28 or B7, both of which are costimulatory signals essential for T-cell activation, attenuated pathological cardiac hypertrophy and HF by reducing the accumulation of activated effector T cells and cardiac inflammation.^{6,7} These findings suggest that T cells are essential mediators in the progression of cardiac dysfunction and HF.

Recent clinical studies have shown that immune checkpoint inhibitors (ICIs) can cause cardiomyopathy without signs of myocarditis.⁸ The occurrence rate of HF in patients receiving a single ICI agent is ≈5.6%, whereas 6.1% of patients who receive combination/sequential ICI therapy develop HF or cardiomyopathy.⁹ In addition, pre-existing cardiovascular conditions significantly increase the risk of HF in patients undergoing ICI therapy.⁹ However, the mechanisms by which ICIs cause adverse cardiovascular events remain elusive.

Previous studies have shown that global *Pdcd1* knockout (KO) BALB/c mice spontaneously develop dilated cardiomyopathy (DCM) due to the accumulation of cardiac troponin I autoantibodies.^{10,11} Of note, these dilated hearts exhibit no apparent signs of inflammation

on histological examination.¹⁰ In contrast, programmed cell death-1 (PD-1)-deficient C57BL/6 mice present lupus-like arthritis and glomerulonephritis from 6 months of age.¹² These findings underscore the critical role of PD-1 in maintaining cardiac function, suggesting that PD-1 deficiency may lead to distinct autoimmune diseases, depending on the genetic background. In contrast to gene silencing approaches, administration of anti-PD-1 antibody induced myocardial infiltration of T cells and cardiac dysfunction exclusively in tumor-bearing C57BL/6 mice, without HF phenotype in non-tumor-bearing mice.¹³ Further studies have demonstrated that treatment with anti-PD-1 antibody alone caused significant cardiac dysfunction by activating interleukin-17 signaling in the thymus of C57BL/6 mice but not in BALB/c mice.¹⁴ Single-cell RNA sequencing (RNA-seq) has revealed a significant increase in PD-1 expression in cardiac T cells after 1 week of transverse aortic constriction (TAC) surgery in mice.¹⁵ Moreover, clonal cytotoxic CD8⁺ T cells have been found to significantly expand in the blood of patients with ICI-induced myocarditis, as well as in mouse hearts with ICI-associated myocarditis.¹⁶ In addition, PD-1 is inducibly expressed in macrophages, where its level negatively correlates with proinflammatory macrophage polarization and the phagocytic potency of tumor-associated macrophages against tumors.¹⁷ From the aforementioned findings, we hypothesized that PD-1 in T cells or myeloid cells may play an essential role in pathological cardiac remodeling and HF.

To test this hypothesis, anti-PD-1 antibody, T cell-, myeloid-, and CD8⁺ T cell-specific *Pdcd1* deletion, *Cxcr3* and *Gzmb* KO mice were used. We show that administration of anti-PD-1 antibody, T cell- or CD8⁺ T cell-specific *Pdcd1* KO exacerbates TAC-induced cardiac remodeling and HF by enhancing CXCL9/CXCL10-CXCR3 chemotaxis. Consequently, myocardial infiltration of CXCR3⁺ CD8⁺ T cells is augmented, leading to GZMB/perforin-mediated inhibition of NADH dehydrogenase (ubiquinone) flavoprotein 1 (NDUFV1) expression, impairment of mitochondrial complex I function, and cardiomyocyte apoptosis. This study may provide a potential therapeutic target for the prevention of ICI-associated cardiac remodeling and HF.

METHODS

The data, analytical methods, and study materials are available to other researchers for purposes of reproducing the results or replicating the procedure on reasonable request from a corresponding author (Z.-R.Z.). Human samples were obtained from patients at Harbin Medical University Cancer Hospital (approval No. KY2022-15), Chongqing University Cancer Hospital (approval No. CZLS2023085-A), and The Second Hospital of Tianjin Medical University (approval No. KY2023051-01) with written informed consent; approved by the ethics committees of the aforementioned hospitals; and executed conforming to the principles of the Declaration of Helsinki. All animal studies followed the guidelines of the animal care and use committee (approval No. KY2022152) of The First Affiliated Hospital of Harbin Medical University. Detailed methods are provided in the [Supplemental Material](#).

Data Availability

The data that support the findings of this study are available from the corresponding author on reasonable request. Detailed experimental methods and the Major Resources Table are available in the [Supplemental Material](#). Sequencing data have been deposited in the Gene Expression Omnibus (GEO) under accession Nos. GSE327655 and GSE328788. The complete R code to ensure reproducibility is available at Zenodo (<https://doi.org/10.5281/zenodo.20363682>). Previously published data sets reanalyzed in this study were obtained from Gene Expression Omnibus through accession code GSE141910.^{18,19}

Statistical Analysis

Data presented in this study are representative of at least 3 independent experiments and are expressed as mean±SEM. The exact group size (n) for each experiment is specified, with “n” denoting biological replicates, not technical replicates. Biological replicate numbers for each figure can be found in the accompanying figure legend. When normal distribution was satisfied, differences between 2 groups were evaluated with 2-tailed unpaired Student *t* test. If unequal variances were detected through an F test, the Welch *t* test was used. For multiple-group comparisons with 1 variable, 1-way ANOVA was conducted followed by the Tukey post hoc multiple-comparisons test when normal distribution was satisfied. In

cases in which unequal variances were identified with a Brown-Forsythe test, a Brown-Forsythe and Welch ANOVA with the Dunnett T3 multiple-comparisons test were implemented. For multiple-group comparisons with ≥2 variables, 2-way ANOVA was conducted followed by the Sidak multiple-comparison test. For nonnormally distributed data, the Mann-Whitney test (unpaired 2 tailed) was used for 2 groups, and the Kruskal-Wallis test followed by the Dunn test was used for multiple groups. All data analyses were performed with GraphPad Prism 8.4.2 software (San Diego, CA). Statistical significance was set at *P*<0.05, *P*<0.01, or *P*<0.001.

RESULTS

Administration of anti-PD-1 Antibody or T Cell-Specific, but Not Myeloid-Specific, *Pdcd1* KO Exacerbates TAC-Induced Cardiac Remodeling and HF

We analyzed the data extracted from the US FDA Adverse Event Reporting System (January 2015–December 2025) and found a total of 96 159 reported adverse events associated with the use of anti-PD-1 antibodies, including nivolumab, pembrolizumab, and cemiplimab. After disproportionality analyses and multivariable logistic regression adjusted for age and sex and exclusion of other types of cardiotoxicities, including myocarditis, concomitant use of antihypertensive and antidiabetic medications was found to be independently associated with cardiac dysfunction/remodeling-related adverse events ([Tables S1 and S2](#); details described in the [Supplemental Material](#)). These results suggest that preexisting hypertension or diabetes may be associated with anti-PD-1 antibody therapy-increased risk of HF- and cardiac remodeling-related adverse events. Therefore, to investigate the detrimental effects of anti-PD-1 antibody on TAC-induced cardiac remodeling and HF, wild-type C57BL/6 mice were subjected to sham or TAC surgery for 2 weeks followed by intraperitoneal injections of either anti-immunoglobulin G (IgG) antibody (10 mg/kg, control) or anti-PD-1 antibody (10 mg/kg, administered every 3 days) for 3 weeks ([Figure S1A](#)). Although administration of anti-IgG antibody or anti-PD-1 antibody alone did not affect cardiac function or cardiac morphology, anti-PD-1 antibody led to further reductions of left ventricular (LV) ejection fraction and fractional shortening and thickened LV wall in TAC mice ([Figure 1A and Table S3](#)). In addition, TAC-induced increases in heart weight (HW)/body weight, HW/tibial length, cardiomyocyte cross-sectional area, and LV collagen-positive area were further augmented by anti-PD-1 antibody ([Figure 1B–1D and Figure S1B and S1C](#)). Consistently, TAC-induced elevations of hypertrophic genes, including *Nppa*, *Nppb*, and *Myh7*, and fibrotic genes, including *Col1a1*, *Col3a1*, and *Mmp9*, were further

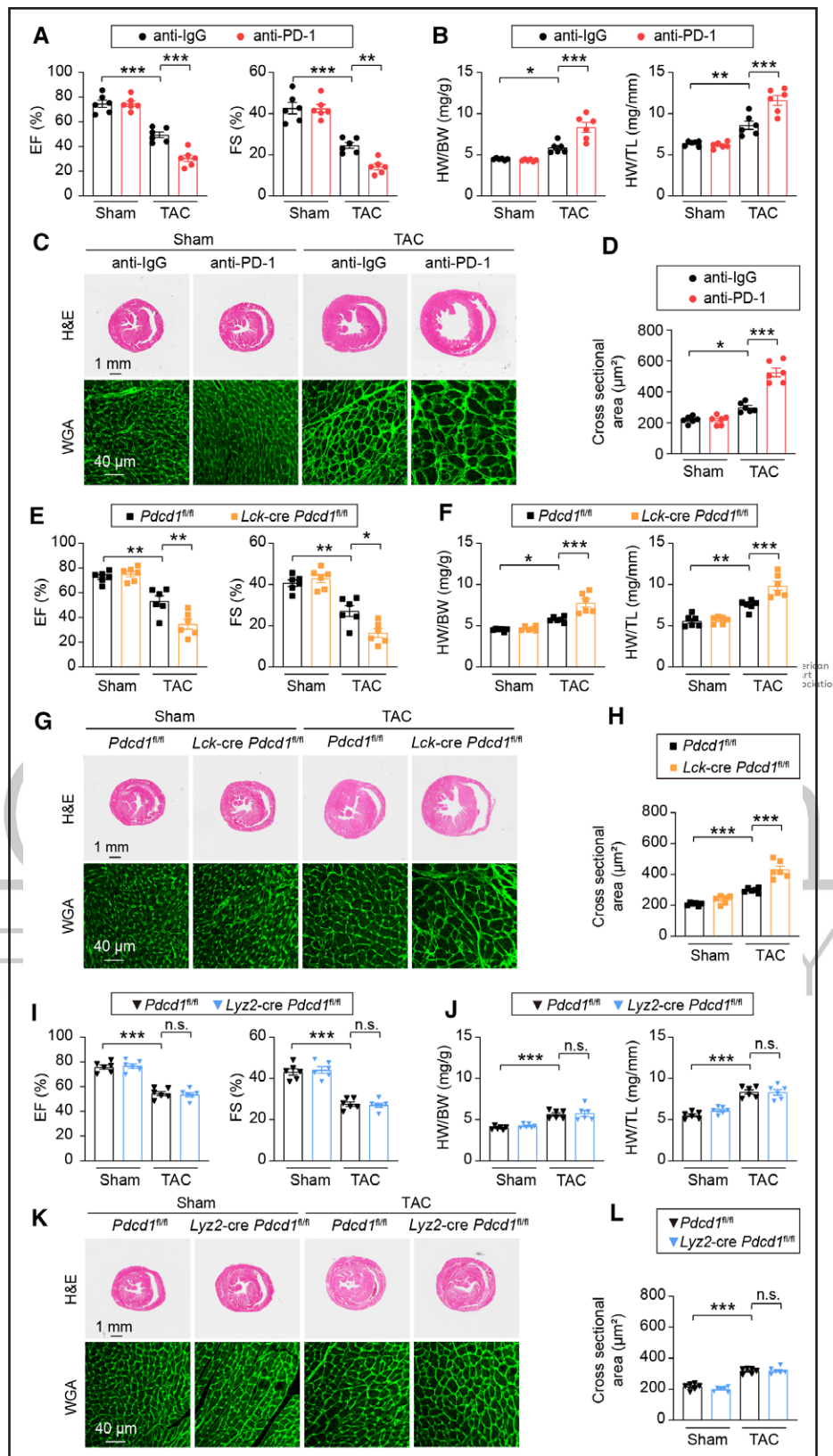


Figure 1. Anti-PD-1 antibody or T cell-specific *Pdc1* deletion exacerbates TAC-induced cardiac remodeling and HF.

A, Cardiac ejection fraction (EF) and fractional shortening (FS) for anti-immunoglobulin G (IgG) antibody- and anti-programmed cell death-1 (PD-1) antibody-treated mice at 5 weeks after sham or transverse aortic constriction (TAC) surgery (n=6 per group). **B,** Heart weight (HW)/body weight (BW) and HW/tibia length (TL) ratios of mice from the indicated experimental groups (n=6 per group). **C,** Representative images of heart sections with hematoxylin and eosin (H&E) staining and wheat germ agglutinin (WGA) staining. **D,** Quantification of the (Continued)

Figure 1 Continued. cross-sectional areas of cardiomyocytes (50 cardiomyocytes per field were evaluated, $n=6$ per group). **E**, Cardiac EF and FS for T cell-specific *Pdcd1* knockout (KO) mice and littermate controls at 5 weeks after sham or TAC surgery ($n=6$ per group). **F**, HW/BW and HW/TL ratios of mice from the indicated experimental groups ($n=6$ per group). **G**, Representative images of heart sections with H&E staining and WGA staining. **H**, Quantification of the cross-sectional areas of cardiomyocytes (50 cardiomyocytes per field were evaluated, $n=6$ per group). **I**, Cardiac EF and FS for myeloid-specific *Pdcd1* KO mice and littermate controls at 5 weeks after sham or TAC surgery ($n=6$ per group). **J**, HW/BW and HW/TL ratios of mice from the indicated experimental groups ($n=6$ per group). **K**, Representative images of heart sections with H&E staining and WGA staining. **L**, Quantification of the cross-sectional areas of cardiomyocytes (50 cardiomyocytes per field were evaluated, $n=6$ per group). Scale bars, 1 mm for H&E staining and 40 μ m for WGA staining. Data are presented as mean $\pm$ SEM. Statistics: Two-way ANOVA, followed by Sidak post hoc multiple comparisons tests (**A**, **B**, **D** through **F**, **H** through **J**, and **L**). HF indicates heart failure; and n.s., no significance between the 2 indicated groups. * $P<0.05$. ** $P<0.01$. *** $P<0.001$.

elevated by anti-PD-1 antibody in mouse LVs (Figure S1D–S1L). These data together indicate that administration of anti-PD-1 antibody significantly exacerbates the severity and progression of TAC-induced cardiac dysfunction.

PD-1 is widely expressed on immune cells. We recently revealed that macrophage PD-1 regulates energy expenditure and systemic metabolic dysfunction under PD-1 blockade.²⁰ Therefore, to identify which specific immune cell is involved in the detrimental effects of anti-PD-1 antibody on TAC-induced cardiac dysfunction, we generated T cell-specific *Pdcd1* KO (*Lck*-cre; *Pdcd1*^{fl/fl}), CD8⁺ T cell-specific *Pdcd1* KO (*Cd8 α* -cre; *Pdcd1*^{fl/fl}), and myeloid-specific *Pdcd1* KO (*Lyz2*-cre; *Pdcd1*^{fl/fl}) mice on the C57BL/6 background by crossing *Pdcd1*-floxed (*Pdcd1*^{fl/fl}) mice with *Lck*-cre, *Cd8 α* -cre, and *Lyz2*-cre mice, respectively (Figure S2A–S2F). Of note, T cell-specific *Pdcd1* KO mice exhibited no signs of cardiac functional or morphological impairment at baseline, indicating that genetic deletion of PD-1 in T cells does not affect cardiac development (Figure 1E–1H and Figure S3A–S3L). Next, T cell-specific *Pdcd1* KO mice and their littermate controls were subjected to either sham or TAC surgery (Figure S3A). Of note, TAC-induced cardiomyopathy, including decreases in LV ejection fraction and fractional shortening and increases in the ratios of HW/body weight and HW/tibial length, was more severe in T cell-specific *Pdcd1* KO mice compared to those in littermate controls (Figure 1E and 1F and Table S4). The hypertrophic phenotypes of T cell-specific *Pdcd1* KO mice subjected to TAC were comparable to those observed in anti-PD-1 antibody-treated TAC mice (Figure 1G and 1H and Figure S3B–S3L). In contrast, myeloid-specific *Pdcd1* KO did not affect TAC-induced hypertrophic phenotypes (Figure 1I–1L; Figure S3A, S3M, and S3N; and Table S5). In addition, neither anti-PD-1 antibody nor T cell-specific *Pdcd1* KO led to structural abnormalities in the kidney, spleen, muscle, liver, lung, or intestine, as revealed by hematoxylin and eosin staining (Figure S4A and S4B).

To investigate whether anti-PD-1 antibody induces additive cardiotoxicity in T cell-specific *Pdcd1* KO mice, anti-PD-1 antibody or anti-IgG antibody was administered to T cell-specific *Pdcd1* KO mice and littermate controls 2 weeks after TAC surgery (Figure S5A). Consistently, both anti-PD-1 antibody and T cell-specific

Pdcd1 KO exacerbated TAC-induced cardiomyopathy, as evidenced by impaired cardiac function, and further augmented hypertrophy and fibrosis (Figure S5B–S5J and Table S6). Of note, anti-PD-1 antibody did not further aggravate cardiac injury in TAC-subjected T cell-specific *Pdcd1* KO mice (Figure S5B–S5J and Table S6), suggesting that anti-PD-1 antibody accelerates TAC-induced progression of cardiac remodeling and HF by targeting PD-1 expressed on T cells.

Suppressing PD-1 Enhances Myocardial Infiltration of CD8⁺ T Cells in TAC Mice

Flow cytometry and immunofluorescence staining were used to identify which specific type of infiltrated immune cells contributed to anti-PD-1 antibody-induced worsening of cardiac remodeling and HF in TAC mice. Our data showed that myocardial infiltration of CD4⁺ T cells, CD8⁺ T cells, and GZMB⁺ CD8⁺ T cells in the LVs of TAC mice was significantly increased by either anti-PD-1 antibody treatment or T cell-specific *Pdcd1* KO (Figure 2A–2N and Figures S6A and S6B and S7A–S7D). Of note, GZMB was not detected in CD4⁺ T cells (Figure S7E and S7F). In contrast, neither anti-PD-1 antibody treatment nor T cell-specific *Pdcd1* KO altered myocardial infiltration of neutrophils, proinflammatory monocytes, or macrophages in TAC mouse hearts (Figure S6C–S6K). Consistently, either anti-PD-1 antibody or T cell-specific *Pdcd1* KO further increased T-cell marker *Cd3e* mRNA and cytotoxic effector molecule *Gzmb* and *Prf1* mRNA expression in the LVs of TAC mice (Figure 2O–2T). The protein levels of GZMB and perforin (determined by ELISA) and the enzyme activity of GZMB (measured by colorimetric assay) in the LVs of TAC mice were enhanced by anti-PD-1 antibody treatment or T cell-specific *Pdcd1* KO (Figure 2U–2Z). Next, CD8⁺ T cell-specific *Pdcd1* KO mice and littermate controls were subjected to either sham or TAC surgery for 5 weeks (Figure S8A). TAC resulted in more severe cardiac injury in CD8⁺ T cell-specific *Pdcd1* KO mice compared with their littermate controls, as evidenced by decreases in LV ejection fraction and fractional shortening, increases in the ratios of HW/body weight and HW/tibial length, and exacerbated cardiac hypertrophy and fibrosis (Figure 3A–3D, Figure S8B–S8L, and Table S7). Moreover, CD8⁺ T cell-specific *Pdcd1* KO further

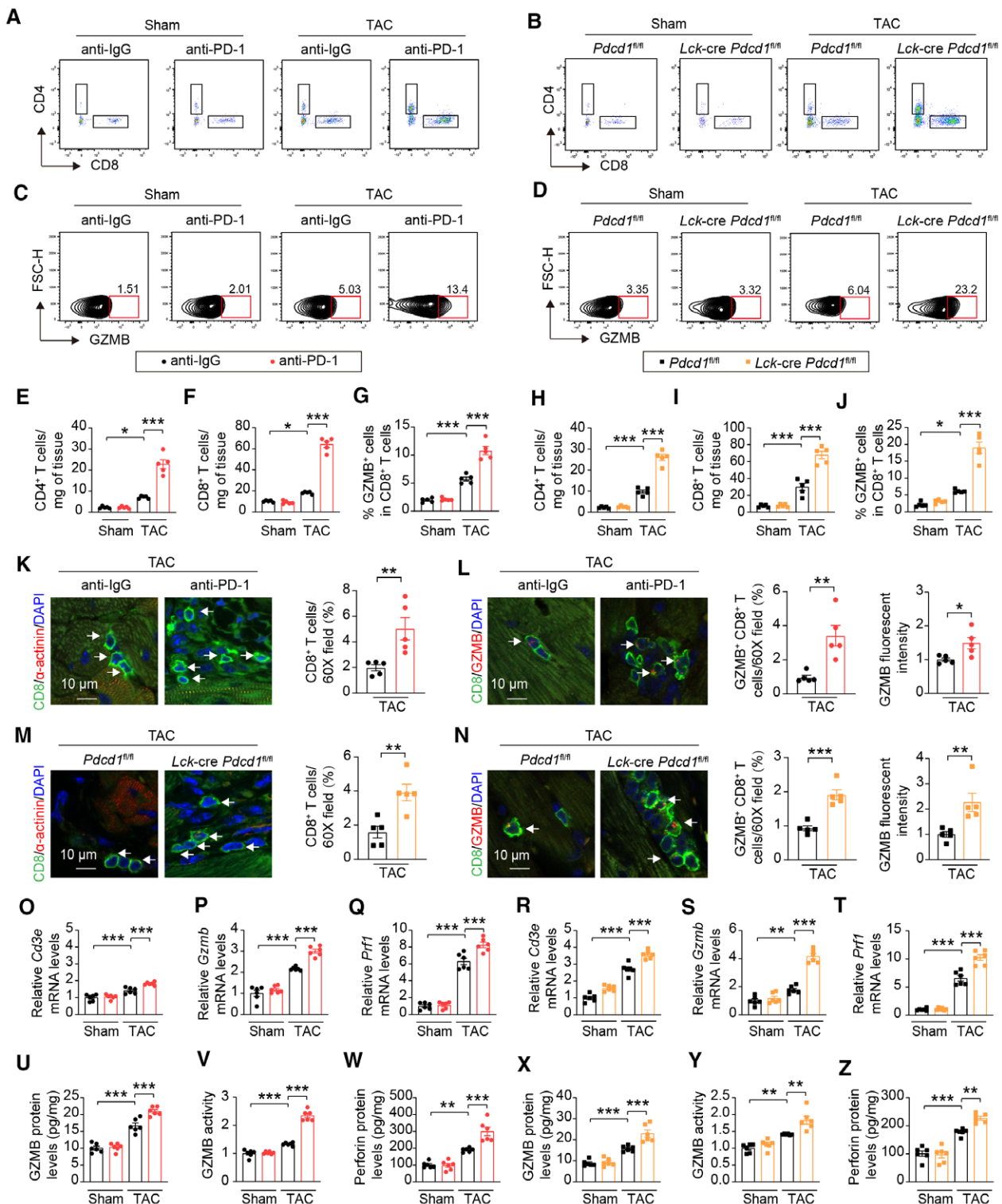


Figure 2. PD-1 blockade or T cell-specific *Pcd1* deletion augments TAC-induced myocardial cytotoxic GZMB⁺ CD8⁺ T cell infiltration.

A through J. Flow cytometry plots and quantitative data, obtained from the cardiac tissue of anti-programmed cell death-1 (PD-1) antibody treatment or T cell-specific *Pcd1* knockout (KO) mice groups, demonstrating the number of CD45⁺ CD3⁺ CD4⁺ T cells, CD45⁺ CD3⁺ CD8⁺ T cells per 1 mg left ventricle (LV), and the percentage of granzyme B (GZMB)⁺ CD8⁺ T cells in CD8⁺ T cells (n=5 per group). **K through N.** Representative confocal microscopy images and quantification of CD8⁺ cells per field, GZMB⁺ CD8⁺ T cells per field, and GZMB fluorescent intensity in heart sections. Immunofluorescence staining was performed with anti- α -actinin antibody (red) to label cardiomyocytes, anti-CD8 antibody (green) to detect CD8⁺ T cells, and anti-GZMB antibody (red) to identify GZMB⁺ cells. Nuclei were labeled with (Continued)

Figure 2 Continued. DAPI (blue). White arrows indicate CD8⁺ T cells and GZMB⁺ CD8⁺ T cells (n=5 per group). **O** through **T**, Relative mRNA expression levels of *Cd3e*, *Gzmb*, and *Prf1* in the LVs of mice from the indicated experimental groups (n=6 per group). **U** through **Z**, Cardiac GZMB and perforin protein levels were determined by ELISA, with GZMB enzymatic activity assessed by colorimetric assay (n=6 per group). Scale bar, 10 μ m. Data are presented as mean $\pm$ SEM. Statistics: unpaired *t* test (**K**, **M**, and **N** for GZMB⁺ CD8⁺ T cells); Mann-Whitney test (**L** for GZMB⁺ CD8⁺ T cells and **N** for GZMB fluorescent intensity); unpaired *t* test with the Welch correction (**L** for GZMB fluorescent intensity); 2-way ANOVA followed by the Sidak post hoc multiple-comparisons tests (**E** through **J**, and **O** through **Z**). TAC indicates transverse aortic constriction. **P*<0.05. ***P*<0.01. ****P*<0.001.

increased myocardial infiltration of CD8⁺ T cells and GZMB⁺ CD8⁺ T cells, the fluorescent intensity of GZMB (Figure 3E–3H), the mRNA levels of *Cd3e*, and the effector molecules *Gzmb* and *Prf1* expression in the LVs of TAC mice (Figure 3I–3K). Collectively, these data indicate that enhanced myocardial infiltration of CD8⁺ T cells contributes to PD-1 blockade-induced worsening of cardiac remodeling and HF in TAC mice.

Suppression of PD-1 Increases TAC-Induced Myocardial Infiltration of Cytotoxic CD8⁺ T Cells Through CXCL9/CXCL10-CXCR3 Chemotaxis

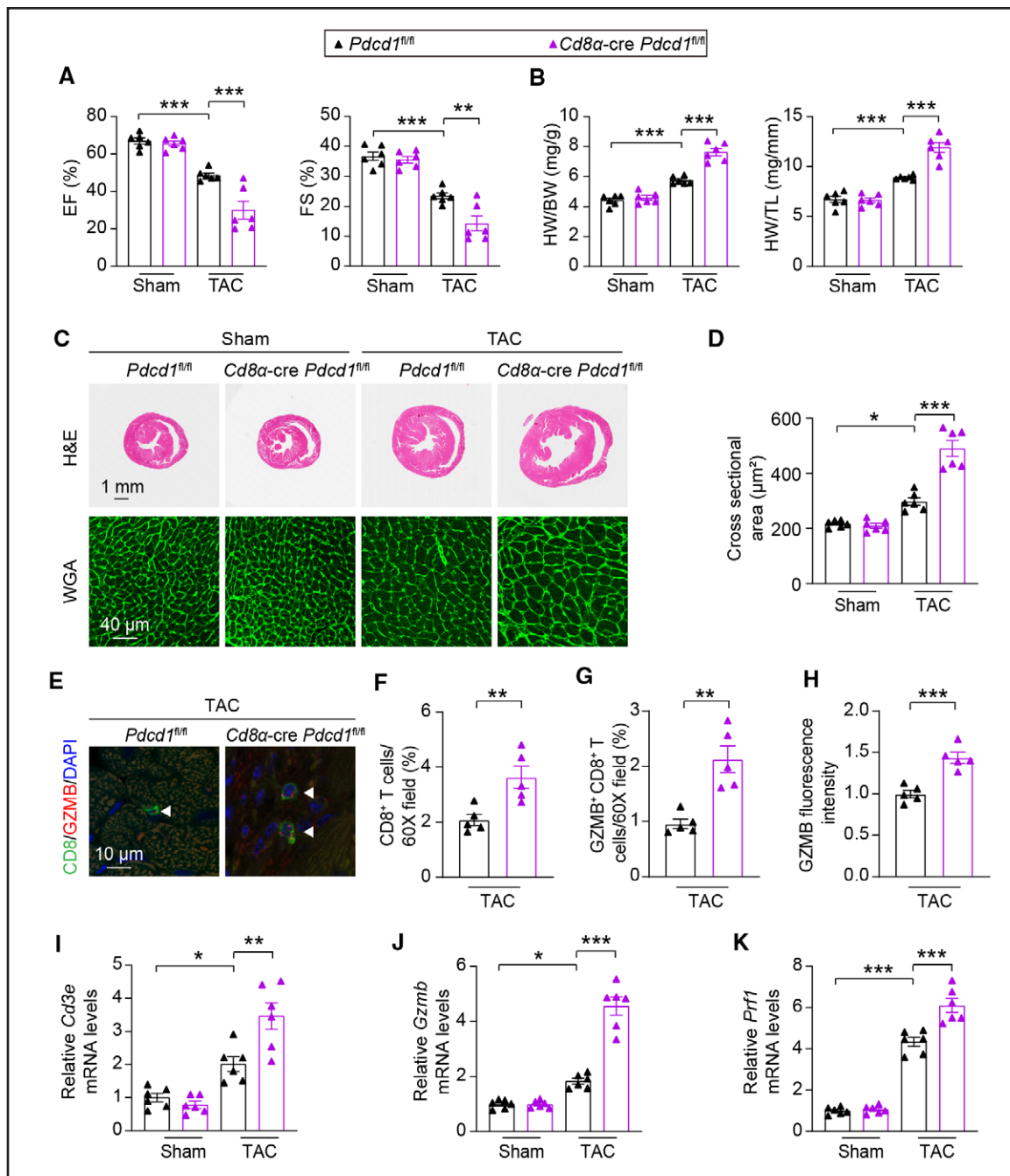
Tissue infiltration of T cells is mediated by the chemoattractant pairs between CXCR3 and CXCL9/CXCL10, CCR5 and CCL3/CCL4/CCL5, and CXCR6 and CXCL16.^{21–23} Of note, expression of these chemoattractant pairs was significantly upregulated in the LVs of TAC mice (Figures S9A–S9R and S10A–S10L). TAC-induced upregulation of *Cxcl9/Cxcl10* and *Cxcr3* and myocardial infiltration of CXCR3⁺ CD8⁺ T cells were further increased by anti-PD-1 antibody or T cell- or CD8⁺ T cell-specific *Pdcd1* KO (Figure S9A–S9R). However, TAC-induced increases in the expression of *Ccr5* and its ligands (*Ccl3*, *Ccl4*, *Ccl5*) and *Cxcr6/Cxcl16* were not affected by anti-PD-1 antibody or T cell-specific *Pdcd1* KO (Figure S10A–S10L).

We then explored the cardiac origin of CXCL9/CXCL10 using cultured neonatal mouse cardiomyocytes (NMCMs) and cardiac fibroblasts (CFs), with or without angiotensin II (Ang II) stimulation. Ang II treatment significantly increased the mRNA levels of *Cxcl9/Cxcl10* in CFs but not in NMCMs. Of note, Ang II-induced upregulation of *Cxcl9/Cxcl10* in CFs was blocked by either AT1 receptor antagonist losartan or nuclear factor- κ B inhibitor BAY 11-7082 (Figure S11A–S11D). Moreover, RNA-seq data from GSE141910,^{18,19} comprising samples of human hypertrophic cardiomyopathy, DCM, and nonfailing LVs, also demonstrated that CXCL9/CXCL10 mRNA expression levels were significantly increased in the LVs of patients with hypertrophic cardiomyopathy or DCM compared with those with nonfailing myocardium (Figure S11E and S11F). These data demonstrate that TAC-induced renin-angiotensin system activation stimulates CXCL9/CXCL10 secretion by CFs, thereby promoting CD8⁺ T cell myocardial infiltration through CXCR3 signaling.

Suppressing CXCR3 Ameliorates PD-1 Blockade- or T Cell-Specific *Pdcd1* Deletion-Induced Worsening of Cardiac Remodeling and HF in TAC Mice

We next used AMG487, a selective CXCR3 antagonist,^{24–26} to explore the role of CXCR3 in the exacerbating effects of PD-1 inhibition on TAC-induced cardiac remodeling in mice (Figures S12A and S13A). Notably, AMG487 prevented the development of cardiac remodeling and HF in TAC mice and abolished PD-1 blockade- or T cell-specific *Pdcd1* deletion-induced worsening of cardiomyopathy (Figure 4A–4H, Figures S12B–S12L and S13B–S13L, and Tables S8 and S9). To further validate the role of CXCR3 in regulating PD-1 blockade-induced exacerbation of TAC-induced cardiac remodeling and HF, we generated global *Cxcr3* KO mice (Figure S14A–S14D). Anti-PD-1 antibody or anti-IgG antibody treatment was initiated at 2 weeks after sham or TAC surgery in *Cxcr3* KO mice and littermate controls (Figure S15A). CXCR3 deficiency not only attenuated baseline TAC-induced cardiac dysfunction but also abrogated anti-PD-1 antibody-induced worsening of cardiac remodeling and HF, as evidenced by improved cardiac function and attenuated hypertrophic phenotypes (Figure 4I–4L, Figure S15B–S15L, and Table S10).

Furthermore, anti-PD-1 antibody- or T cell-specific *Pdcd1* KO-mediated worsening effects on the myocardial infiltration of CD8⁺ T cells and the increased expression levels of *Cxcl9/Cxcl10*, *Cxcr3*, *Cd3e*, *Gzmb*, and *Prf1* in the LVs of TAC mice were dramatically inhibited by AMG487 (Figure 5A–5N). Consistently, the TAC-induced phenomena observed in anti-PD-1 antibody-treated mice were abolished in *Cxcr3* KO mice (Figure 5O–5U). To further answer how AMG487 ameliorates anti-PD-1 antibody-induced worsening of cardiac remodeling and HF in TAC mice, we performed *in vitro* transwell assays using CD8⁺ T cells isolated from *Cxcr3* KO mice or littermate controls, with or without CXCL9/CXCL10 treatment. The chemotaxis of CD8⁺ T cells toward CXCL9/CXCL10 was markedly inhibited by either CXCR3 antagonist AMG487 or *Cxcr3* KO. However, AMG487 did not exhibit any additional inhibitory effect on the migration of CD8⁺ T cells isolated from *Cxcr3* KO mice (Figure S16A). In addition, the expression levels of GZMB and perforin were not altered by *Cxcr3* KO, AMG487, or their combination (Figure S16B–S16D), suggesting that CXCR3 regulates CD8⁺ T cell



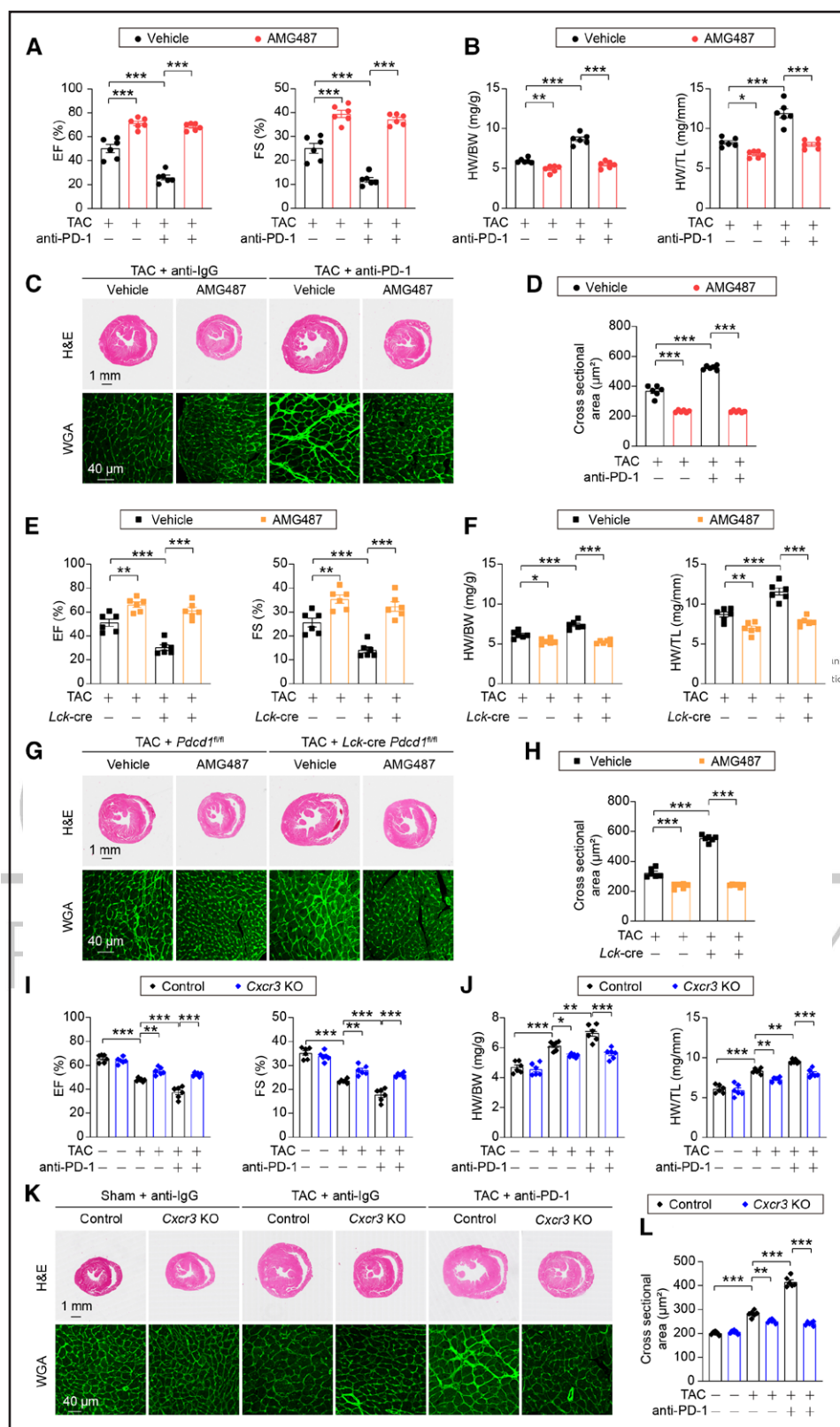


Figure 4. CXCR3 antagonist AMG487 or *Cxcr3* KO ameliorates PD-1 blockade- or T cell-specific *Pdc1* deletion-induced exacerbation of TAC-induced cardiac remodeling and HF.

A, Cardiac ejection fraction (EF) and fractional shortening (FS) for anti-immunoglobulin G (IgG) antibody- or anti-programmed cell death-1 (PD-1) antibody-treated wild-type (WT) mice in the presence of AMG487 or vehicle for 3 weeks at 5 weeks after transverse aortic constriction (TAC) surgery (n=6 per group). **B**, Heart weight (HW)/body weight (BW) and HW/tibia length (TL) ratios of mice from the indicated experimental groups (n=6 per group). **C**, Representative images of heart sections with hematoxylin and eosin (H&E) staining and wheat germ (Continued)

Figure 4 Continued. agglutinin (WGA) staining. **D**, Quantification of the cross-sectional areas of cardiomyocytes (50 cardiomyocytes per field were evaluated, n=6 per group). **E**, Cardiac EF and FS for T cell-specific *Pdcd1* knockout (KO) mice and littermate controls in the presence of AMG487 or vehicle for 5 weeks at 5 weeks of TAC surgery (n=6 per group). **F**, HW/BW and HW/TL ratios of mice from the indicated experimental groups (n=6 per group). **G**, Representative images of heart sections with H&E staining and WGA staining. **H**, Quantification of the cross-sectional areas of cardiomyocytes (50 cardiomyocytes per field were evaluated, n=6 per group). **I**, Cardiac EF and FS for *Cxcr3* KO mice and littermate controls at 5 weeks after sham or TAC surgery. Mice received either anti-IgG antibody or anti-PD-1 antibody treatment beginning 2 weeks after TAC surgery (n=6 per group). **J**, HW/BW and HW/TL ratios of mice from the indicated experimental groups (n=6 per group). **K**, Representative images of heart sections with H&E staining and WGA staining. **L**, Quantification of the cross-sectional areas of cardiomyocytes (50 cardiomyocytes per field were evaluated, n=6 per group). Scale bars, 1 mm for H&E staining and 40 μ m for WGA staining. Data are presented as mean $\pm$ SEM. Statistics: 2-way ANOVA followed by the Sidak post hoc multiple-comparisons tests (**A**, **B**, **D** through **F**, **H** through **J**, and **L**). HF indicates heart failure. * P <0.05. ** P <0.01. *** P <0.001.

chemotaxis but not cytotoxic effector molecule expression. Collectively, these data suggest that the CXCL9/CXCL10-CXCR3 chemotaxis-mediated enhancement of GZMB⁺ CD8⁺ T-cell myocardial infiltration may play a key role in PD-1 blockade- or T cell-specific *Pdcd1* deletion-induced aggravation of cardiac remodeling and HF under TAC.

PD-1 Blockade or T Cell-Specific *Pdcd1* Deletion Exacerbates TAC-Induced Impairment of Mitochondrial Respiratory Chain Complex I and Cardiomyocyte Apoptosis

We next analyzed the effect of PD-1 inhibition on gene expression of TAC hearts by RNA-seq. Compared with anti-IgG antibody-treated TAC mice, 517 and 886 differentially expressed genes were upregulated and downregulated, respectively, in the LVs of anti-PD-1 antibody-treated TAC mice (Figure 6A). Gene Ontology analyses revealed that the downregulated biological processes were associated with cellular respiration, oxidative phosphorylation, and ATP biosynthetic process (Figure 6B). Gene set enrichment analysis demonstrated that the upregulated biological processes were related to hypertrophic cardiomyopathy and DCM, whereas downregulated genes were enriched in cardiac muscle contraction and mitochondrial oxidative phosphorylation pathways (Figure S17A–S17D). We further confirmed that the decreased expression of NDUFV1, a subunit of mitochondrial complex I, the increased Bax/Bcl-2 ratio, and the number of TUNEL (TdT-mediated dUTP nick-end labeling)-positive cells were further aggravated by anti-PD-1 antibody treatment, T cell-, or CD8⁺ T cell-specific *Pdcd1* KO (Figures S18A–S18I and S19A–S19F). However, the expression levels of other respiratory chain proteins, including NDUF51/2, ATP5A, UQCRC2, MTCO1, SDHB, and NDUF8, were not affected by PD-1 blockade and T cell-specific *Pdcd1* KO (Figure S20A–S20R). Notably, the aforementioned pathological alterations and the elevated serum levels of NT-proBNP (N-terminal pro-B-type natriuretic peptide) and cardiac troponin I were diminished by AMG487 or *Cxcr3* KO

(Figure 6C–6K and Figures S21A–S21F and S22A–S22F). These findings indicate that deletion or inhibition of PD-1 in CD8⁺ T cells may further exacerbate mitochondrial respiration and cardiomyocyte apoptosis in response to hemodynamic stress, which can be prevented by CXCR3 antagonism or *Cxcr3* KO.

GZMB Acts as a Key Molecule in Exacerbating PD-1 Blockade- or T Cell-Specific *Pdcd1* Deletion-Induced Cardiac Dysfunction and HF Under Pressure Overload



We next used global *Gzmb* KO mice to explore the role of GZMB in the exacerbating effects of PD-1 inhibition on TAC-induced cardiac dysfunction. *Gzmb* KO mice and littermate controls were subjected to either sham or TAC surgery for 2 weeks followed by intraperitoneal injection of anti-IgG antibody or anti-PD-1 antibody for an additional 3 weeks (Figures S23A–S23C and S24A). *Gzmb* deletion significantly ameliorated hypertrophic phenotypes in anti-PD-1 antibody-treated TAC mice (Figure 7A–7D, Figure S24B–S24L, and Table S11). Moreover, *Gzmb* KO significantly attenuated the increase in mRNA levels of *Cxcl9*, *Cxcl10*, *Cxcr3*, *Cd3e*, *Gzmb*, and *Prf1* in the LVs of anti-PD-1 antibody-treated TAC mice (Figure 7E–7J). *Gzmb* deletion also greatly blunted the increased myocardial infiltration of cytotoxic CD8⁺ T cells and GZMB⁺ CD8⁺ T cells (Figure 7K–7N). Notably, GZMB expression in CD8⁺ T cells was almost undetectable in the LVs of GZMB-deficient TAC mice treated with anti-PD-1 antibody (Figure 7K–7N). Of note, the TAC-induced decrease in NDUFV1 expression, the increase in the Bax/Bcl-2 ratio and TUNEL-positive cells, and elevated serum levels of NT-proBNP and cardiac troponin I were also augmented by anti-PD-1 antibody, which were reversed by *Gzmb* KO (Figure 7O–7Q and Figures S24M, S24N, S25A and S25B).

To further validate the role of GZMB in the exacerbating effect of *Pdcd1* deletion on TAC-induced cardiac dysfunction, GZMB-neutralizing antibody or anti-IgG antibody was applied to T cell-specific *Pdcd1* KO mice and littermate controls subjected to sham or TAC surgery for 5 consecutive weeks (Figure S26A). Consistently,

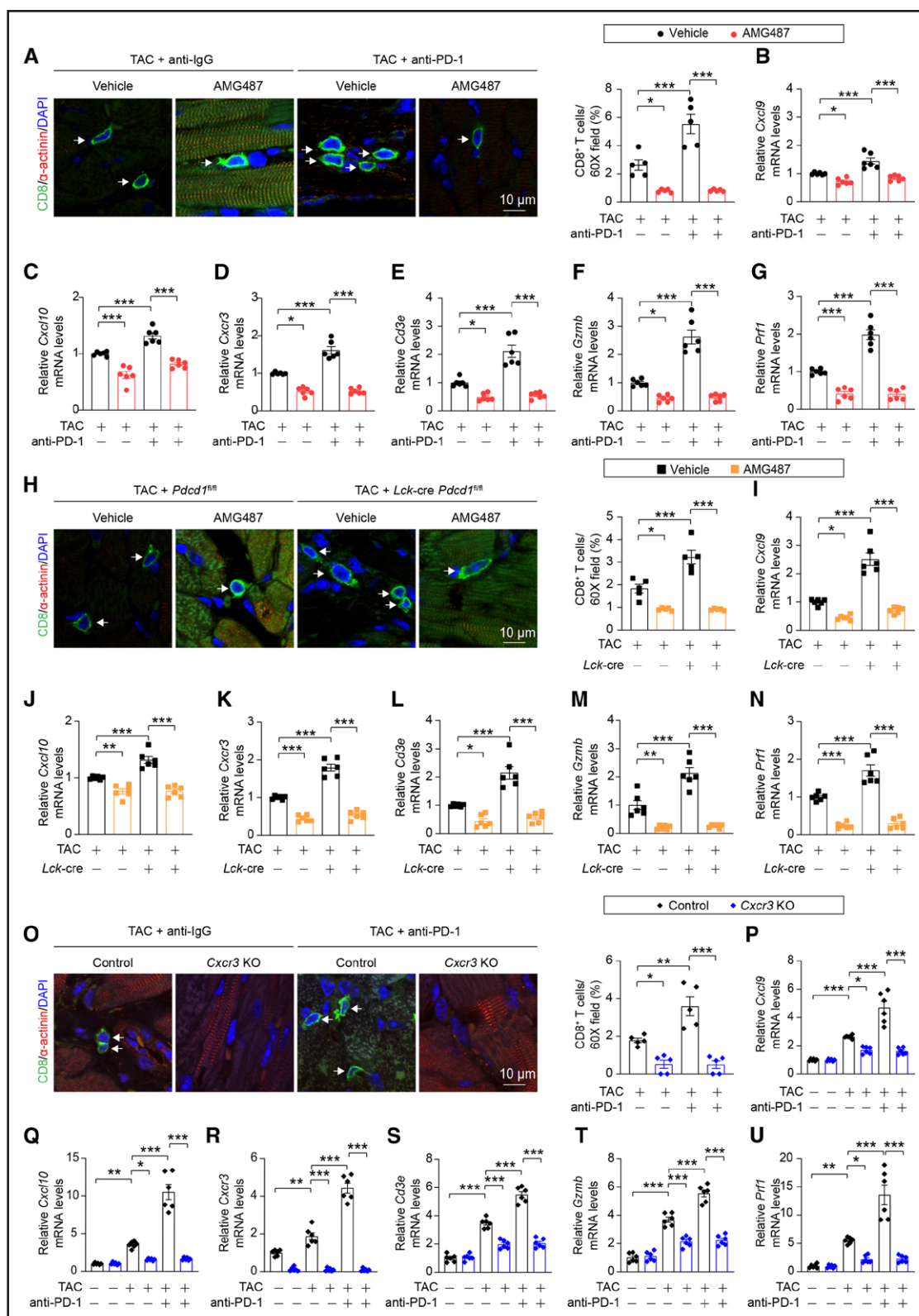


Figure 5. The CXCL9/CXCL10-CXCR3 chemotaxis mediates the myocardial infiltration of GZMB⁺ CD8⁺ T cells in TAC mice.

A through **U**, Representative confocal images and quantification of CD8⁺ T cells per field in heart sections from the indicated experimental groups. Cardiomyocytes were identified with anti- α -actinin antibody (red) staining; CD8⁺ T cells were identified with anti-CD8 antibody (green) staining; and nuclei were labeled with DAPI (blue). White arrows indicate CD8⁺ T cells (**A**, **H**, and **O**; $n=5$ per group). Relative mRNA expression levels of *Cxcl9*, *Cxcl10*, *Cxcr3*, *Cd3e*, *Gzmb*, and *Prf1* in the left ventricles (LVs) of mice (**B** through **G**, **I** through **N**, and **P** through **U**; $n=6$ per group). Scale bar, 10 μ m. Data are presented as mean $\pm$ SEM. Statistics: 2-way ANOVA followed by the Sidak post hoc multiple-comparisons tests (**A** through **U**). GZMB indicates granzyme B; and TAC, transverse aortic constriction. * $P<0.05$. ** $P<0.01$. *** $P<0.001$.

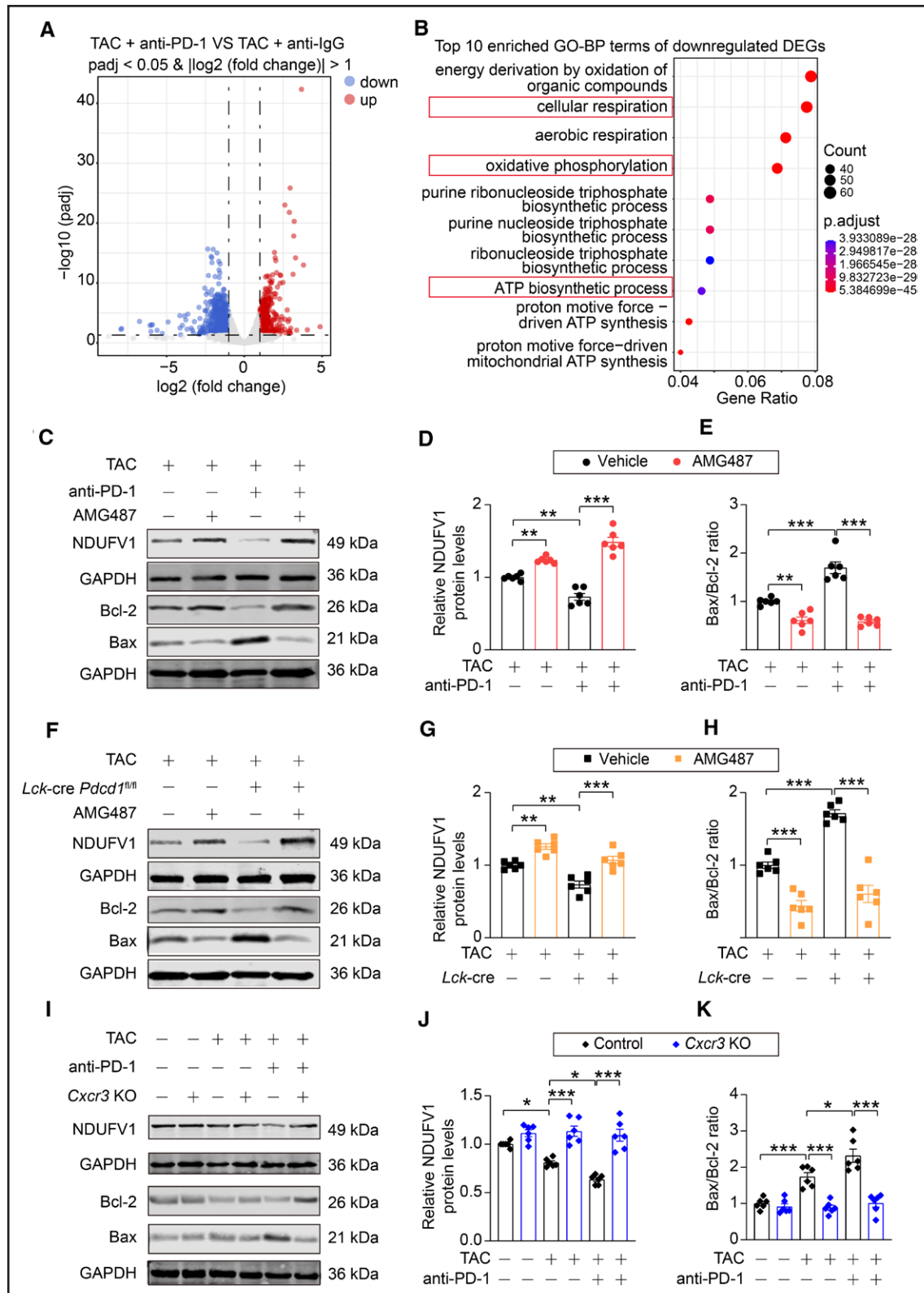


Figure 6. PD-1 blockade- or T cell-specific *Pdc1* deletion-induced exacerbation of mitochondrial respiratory chain complex I dysfunction and apoptosis in cardiomyocytes of TAC mice is prevented by AMG487 or *Cxcr3* KO.

A, Volcano plot showing differentially expressed genes (DEGs; red and blue points indicate upregulated genes and downregulated genes, respectively) in the left ventricles (LVs), obtained from anti-immunoglobulin G (IgG) antibody- and anti-programmed cell death-1 (PD-1) antibody-treated transverse aortic constriction (TAC) mice. Adjusted $P < 0.05$ and average $|\log_2(\text{fold change})| > 1$ were considered significantly regulated by anti-PD-1 antibody ($n = 3$ per group). **B**, Gene Ontology (GO) analysis of downregulated genes in RNA-sequencing data. Dot size represents the number of input genes associated with each GO term. Gene ratio indicates the number of enriched genes divided (Continued)

Figure 6 Continued. by the total genes in that GO term in background. Dot color intensity represents the statistical significance by the adjusted *P* value (Benjamini-Hochberg). All significantly DEGs presented in **B** are provided in [Supplemental Excel Data S1](#). **C** through **K**, Representative Western blots and quantification of NADH dehydrogenase (ubiquinone) flavoprotein 1 (NDUFV1), Bcl-2-associated X (Bax), and B-cell lymphoma-2 (Bcl-2) in the LVs of mice from the indicated experimental groups (n=6 per group). Data are presented as mean±SEM. Statistics: Two-way ANOVA, followed by Sidak post hoc multiple comparisons tests (**D**, **E**, **G**, **H**, **J**, and **K**). KO indicates knockout. **P*<0.05. ***P*<0.01. ****P*<0.001.

GZMB neutralization not only ameliorated TAC-induced cardiac injury in littermate control mice but also rescued the worsening effect of T cell-specific *Pdcd1* KO on TAC-induced pathological cardiac remodeling and HF ([Figure S26B–S26J](#) and [Table S12](#)). These data suggest that GZMB may act as a direct cytotoxic molecule downstream of PD-1.

Increased GZMB and Perforin Cause Cardiomyocyte Apoptosis By Impairing Mitochondrial Complex I Activity

We then tested whether GZMB plays a key role in the worsening of PD-1 blockade- or T cell-specific *Pdcd1* deletion-induced impairment of mitochondrial respiration and apoptosis in cardiomyocytes. T cells were isolated from mouse spleens ([Figure S27A–S27D](#)). Western blot and ELISA analyses showed that GZMB and perforin were almost undetectable in CD4⁺ T cells and their culture supernatant. In addition, T cell-specific *Pdcd1* KO or anti-PD-1 antibody treatment did not increase GZMB and perforin in these CD4⁺ T cells. However, the levels of GZMB and perforin detected in CD8⁺ T cells and their culture supernatant were dramatically increased by either T cell-specific *Pdcd1* KO ([Figure S28A–S28F](#)) or anti-PD-1 antibody treatment ([Figure S28G–S28L](#)). The stimulatory effect of anti-PD-1 antibody on GZMB expression in CD8⁺ T cells was diminished by *Gzmb* KO ([Figure S28G–S28I](#)), suggesting that CD8⁺ T cell-derived GZMB and perforin play a critical role in the worsening effects of anti-PD-1 antibody or T cell-specific *Pdcd1* KO on TAC-induced cardiac injury.

To further investigate whether there is a link between CD8⁺ T cell activation and anti-PD-1 antibody therapy-induced cardiac injury, single-cell RNA-seq analyses of peripheral blood mononuclear cells from patients receiving anti-PD-1 antibody treatment were performed. We analyzed 44 054 cells from patients without signs of cardiac injury (n=7) after anti-PD-1 antibody therapies and 40 106 cells from those diagnosed as having anti-PD-1 antibody therapy-associated “possible myocarditis”²⁷ (n=7; detailed descriptions of clinical data and criteria are shown in the [Supplemental Material](#) and [Table S13](#)). Unsupervised clustering based on canonical markers identified 6 major immune cell clusters, including monocytes, CD8⁺ T cells, CD4⁺ T cells, $\gamma\delta$ T cells, natural killer cells, and B cells ([Figure S29A](#) and [S29B](#)), with comparable proportions between groups ([Figure S29C](#)). Of note, in patients diagnosed as having anti-PD-1 antibody

therapy-associated possible myocarditis, the expression of *GZMB* and *PRF1* was significantly enhanced in cytotoxic CD8⁺ T cells ([Figure S29D](#) and [S29E](#)). Furthermore, the serum levels of GZMB/perforin, NT-proBNP, cardiac troponin I, and creatine kinase-MB were significantly elevated in patients with anti-PD-1 antibody therapy-associated possible myocarditis ([Figure S29F](#); detailed descriptions shown in [Table S14](#)). These data suggest that the elevated expression of GZMB and perforin in cytotoxic CD8⁺ T cells, along with the increased serum levels of GZMB and perforin, may correlate with anti-PD-1 antibody-induced cardiotoxicity in patients.

Because GZMB directly cleaves NDUFV1 and impairs mitochondrial complex I activity,²⁸ we then respectively cocultured CD4⁺ T cells or CD8⁺ T cells (isolated from respective mouse models) with NMCMs. Notably, CD4⁺ T cells, obtained from either T cell-specific *Pdcd1* KO mice or littermate controls, did not affect NDUFV1 expression or the activity of mitochondrial complex I in NMCMs. However, CD8⁺ T cells significantly inhibited NDUFV1 expression and decreased the activity of mitochondrial complex I in NMCMs, which were further exacerbated by *Pdcd1* deletion ([Figure 8A–8C](#)). Moreover, coculture of NMCMs with CD4⁺ T cells and CD8⁺ T cells (isolated from control mice) resulted in decreased intracellular ATP levels, increased cellular and mitochondrial reactive oxygen species (ROS) accumulation, and cardiomyocyte apoptosis, which were further augmented by *Pdcd1* KO in CD8⁺ T cells but not in CD4⁺ T cells ([Figure 8D–8H](#) and [Figure S30A](#)). Consistently, the effects of *Pdcd1* KO on NMCMs were mimicked only when anti-PD-1 antibody was applied to the culture media with CD8⁺ T cells but not with CD4⁺ T cells ([Figure 8I–8P](#) and [Figure S30B](#)), which were reversed by *Gzmb* KO ([Figure 8Q–8X](#) and [Figure S30C](#)).

Notably, GZMB detected by immunofluorescence staining in NMCMs cocultured with CD8⁺ T cells was abolished by perforin inhibitor perforin-IN-2 or by coculturing NMCMs with CD8⁺ T cells isolated from *Gzmb* KO mice. The reduced NDUFV1 expression and subsequent cardiomyocyte apoptosis were rescued by either perforin-IN-2 or *Gzmb* KO, suggesting that perforin mediates CD8⁺ T cell-derived GZMB entry into cardiomyocytes ([Figure S31A–S31D](#)). Consistently, treatment of NMCMs with recombinant GZMB alone affected neither NDUFV1 expression nor mitochondrial respiratory and apoptosis phenotypes. However, incubation of NMCMs with both recombinant perforin and GZMB suppressed NDUFV1 expression, reduced ATP levels, increased

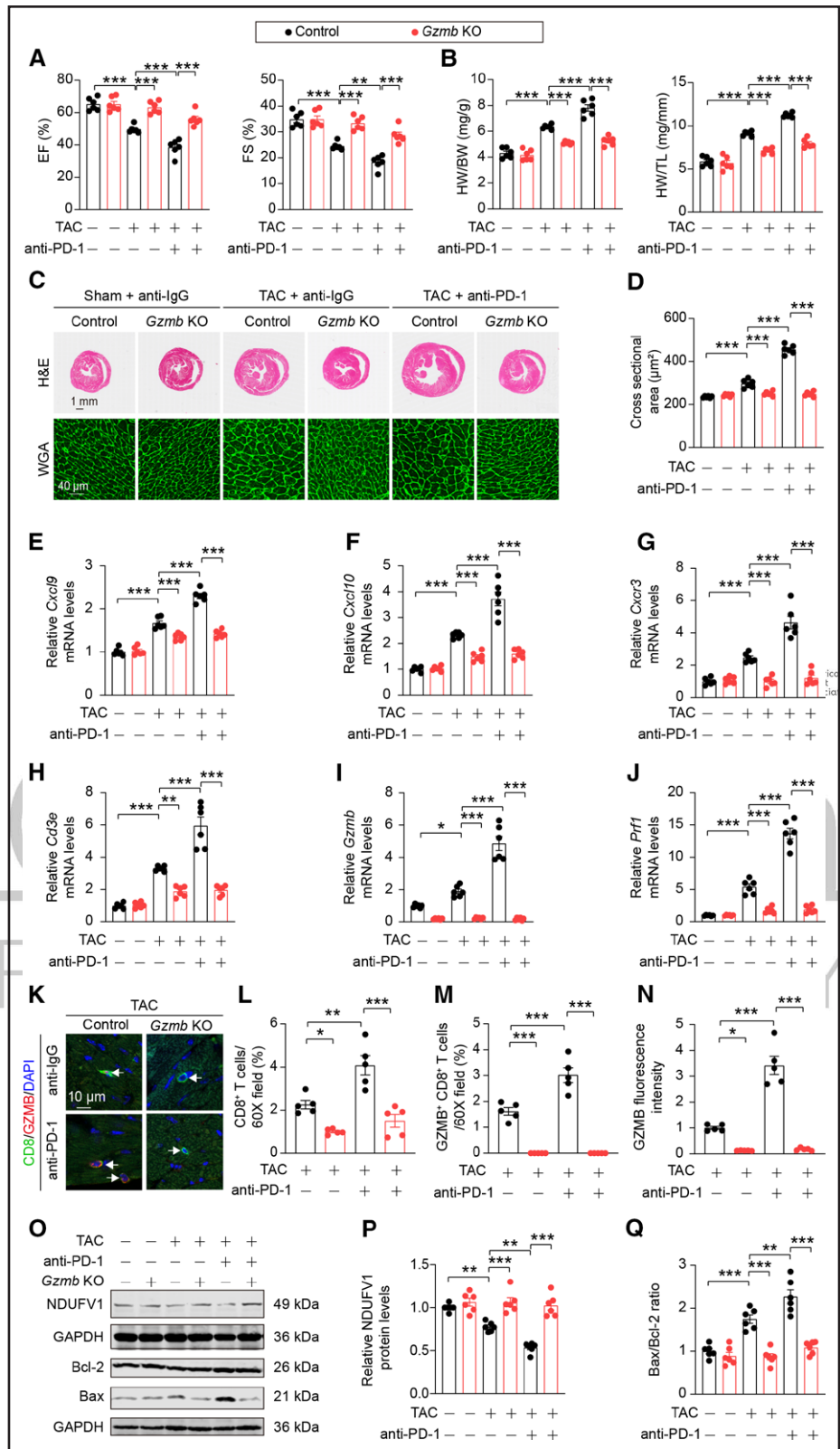


Figure 7. GZMB deletion ameliorates anti-programmed cell death-1 antibody-induced aggravation of cardiac remodeling and HF in TAC mice.

A, Cardiac ejection fraction (EF) and fractional shortening (FS) data obtained from the indicated experimental groups after 5 weeks of sham or transverse aortic constriction (TAC) surgery (n=6 per group). **B**, Heart weight (HW)/body weight (BW) and HW/tibia length (TL) ratios of mice from the indicated experimental groups (n=6 per group). **C**, Representative images of heart sections with hematoxylin and eosin (H&E) staining and wheat germ agglutinin (WGA) staining. Scale bars, 1 mm for H&E staining and 40 μ m for WGA staining. **D**, Quantification (Continued)

Figure 7 Continued. of the cross-sectional areas of cardiomyocytes (50 cardiomyocytes per field were evaluated; n=6 per group). **E** through **J**, Relative mRNA expression levels of *Cxcl9*, *Cxcl10*, *Cxcr3*, *Cd3e*, *Gzmb*, and *Prf1* in the left ventricles (LVs) of mice (n=6 per group). **K** through **N**, Representative confocal microscopy images (**K**) and quantification of CD8⁺ T cells (**L**) per field, granzyme B (GZMB)⁺ CD8⁺ T cells (**M**) per field, and GZMB fluorescent intensity (**N**) in heart sections. Sections were stained with anti-CD8 antibody (green) and anti-GZMB antibody (red) to identify GZMB⁺ CD8⁺ T cells. White arrows indicate CD8⁺ T cells (n=5 per group). **O** through **Q**, Representative Western blots and quantification of NADH dehydrogenase (ubiquinone) flavoprotein 1 (NDUFV1), Bcl-2-associated X (Bax), and B-cell lymphoma-2 (Bcl-2) in the LVs of mice (n=6 per group). Data are presented as mean±SEM. Statistics: 2-way ANOVA followed by the Sidak post hoc multiple-comparisons tests (**A**, **B**, **D** through **J**, **L** through **N**, **P**, and **Q**). HF indicates heart failure. **P*<0.05. ***P*<0.01. ****P*<0.001.

cellular and mitochondrial ROS, induced cardiomyocyte apoptosis, and markedly decreased the activities of complexes I, III, and V, without affecting complexes II or IV. It is important to note that these effects were abolished by the GZMB inhibitor I (Figure S32A–S32M). In addition, application of anti-PD-1 antibody alone did not lead to any adverse effect on NMCMs (Figure S33A–S33I).

Last, to determine whether GZMB could directly exacerbate TAC-induced HF, mice were injected with adeno-associated virus serotype 9 encoding GZMB under the control of the cardiac troponin T promoter (adeno-associated virus serotype 9–cardiac troponin T–GZMB; 5×10^{11} genomic copies per mouse) or adeno-associated virus serotype 9–cardiac troponin T–GFP (green fluorescent protein; served as control) to generate cardiomyocyte-specific GZMB overexpression in vivo. One week after viral injection, mice were subjected to TAC or sham surgery for an additional 4 weeks. The efficiency of cardiomyocyte-specific overexpression of GZMB was confirmed by Western blot (Figure S34A–S34B). Of note, cardiomyocyte-specific GZMB overexpression significantly aggravated TAC-induced cardiac dysfunction, as evidenced by reduced cardiac function, exacerbated cardiac remodeling, and increased expression of fetal gene markers (*Nppa*, *Nppb*, and *Myh7*) and fibrotic markers (*Col1a1*, *Col3a1*, and *Mmp9*; Figures S34C–S34K and S35A–S35F and Table S15). Moreover, cardiomyocyte-specific GZMB overexpression further exacerbated TAC-induced downregulation of NDUFV1, elevated Bax/Bcl-2 ratio, and increased TUNEL-positive cardiomyocytes (Figure S35G–S35K). These data together suggest that PD-1 blockade– or T cell–specific *Pdcd1* deletion–mediated increase in GZMB/perforin expression in CD8⁺ T cells may account for the worsening of TAC-induced hypertrophic phenotypes and HF through impairing mitochondrial respiratory chain complexes I, III, and V and ATP production, excessive ROS production, and cardiomyocyte apoptosis.

DISCUSSION

LV systolic dysfunction and HF are common, delayed cardiac adverse events associated with ICI therapy.²⁹ In patients with preexisting cardiovascular diseases, ICI therapy significantly raises the risk of cardiovascular immune-related adverse events, including HF.³⁰ However, how ICI therapy accelerates HF progression remains incompletely understood. We recently showed that

anti-CTLA-4 m2a antibody did not cause cardiotoxicity in wild-type C57BL/6 mice; however, it exacerbates experimental autoimmune myocarditis-induced cardiac injury by promoting Ccl5-neutrophil myocardial infiltration.³¹ Previous studies showed that application of anti-PD-1 antibody resulted in neither cardiotoxicity in non-tumor-bearing C57BL/6 mice nor inflammation and apoptosis in cardiomyocytes.^{13,32} Consistently, we also found that anti-PD-1 antibody alone did not cause any notable cardiac phenotypes in wild-type C57BL/6 mice or adverse effects on cultured NMCMs. However, TAC-induced cardiac remodeling and HF were dramatically augmented by anti-PD-1 antibody, suggesting that PD-1 may act as a potential regulator in the development of cardiac injury under preexisting pathological conditions.

CD4⁺ T cells, but not CD8⁺ T cells, promote the transition from hypertrophic remodeling to HF in TAC mice.³³ In an autoimmune thyroiditis mouse model, anti-PD-1 antibody damaged the thyroid gland through direct activation of cytotoxic CD4⁺ T cells, and depletion of CD4⁺ T cells reversed anti-PD-1 antibody–induced thyroiditis development.³⁴ Recent studies demonstrated that a significant expansion of clonal cytotoxic CD8⁺ T cells was observed in the peripheral blood of patients with ICI-associated myocarditis compared with those without myocarditis.¹⁶ However, the role of CD4⁺ T cells versus CD8⁺ T cells in mediating anti-PD-1 antibody–aggravated cardiac remodeling and HF during pressure overload remains incompletely understood. We show that both T cell– and myeloid-specific *Pdcd1* KO in C57BL/6 mice did not cause cardiomyopathy under baseline conditions. However, T cell– or CD8⁺ T cell–specific *Pdcd1* KO, but not myeloid-specific *Pdcd1* KO, significantly worsened TAC-induced cardiac remodeling and HF, suggesting that T cells activated by anti-PD-1 antibody account for exacerbating cardiac injury in TAC mice. Of note, anti-PD-1 antibody treatment did not exacerbate TAC-induced cardiac injury in T cell–specific *Pdcd1* KO mice, suggesting that anti-PD-1 antibody–induced cardiac injury is exclusively mediated by PD-1 expressed on T cells.

Anti-PD-1 antibody increased myocardial infiltration of CD4⁺ T cells and CD8⁺ T cells in tumor-bearing mice, in which activated CD8⁺ T cells and tumor necrosis factor- α signaling played key roles in anti-PD-1 antibody therapy–associated cardiotoxicity.¹³ Most recently, an excellent study identified that CXCR6⁺ T cells play a central role in combined therapy of anti-lymphocyte activation gene 3/anti-PD-1 antibody–induced

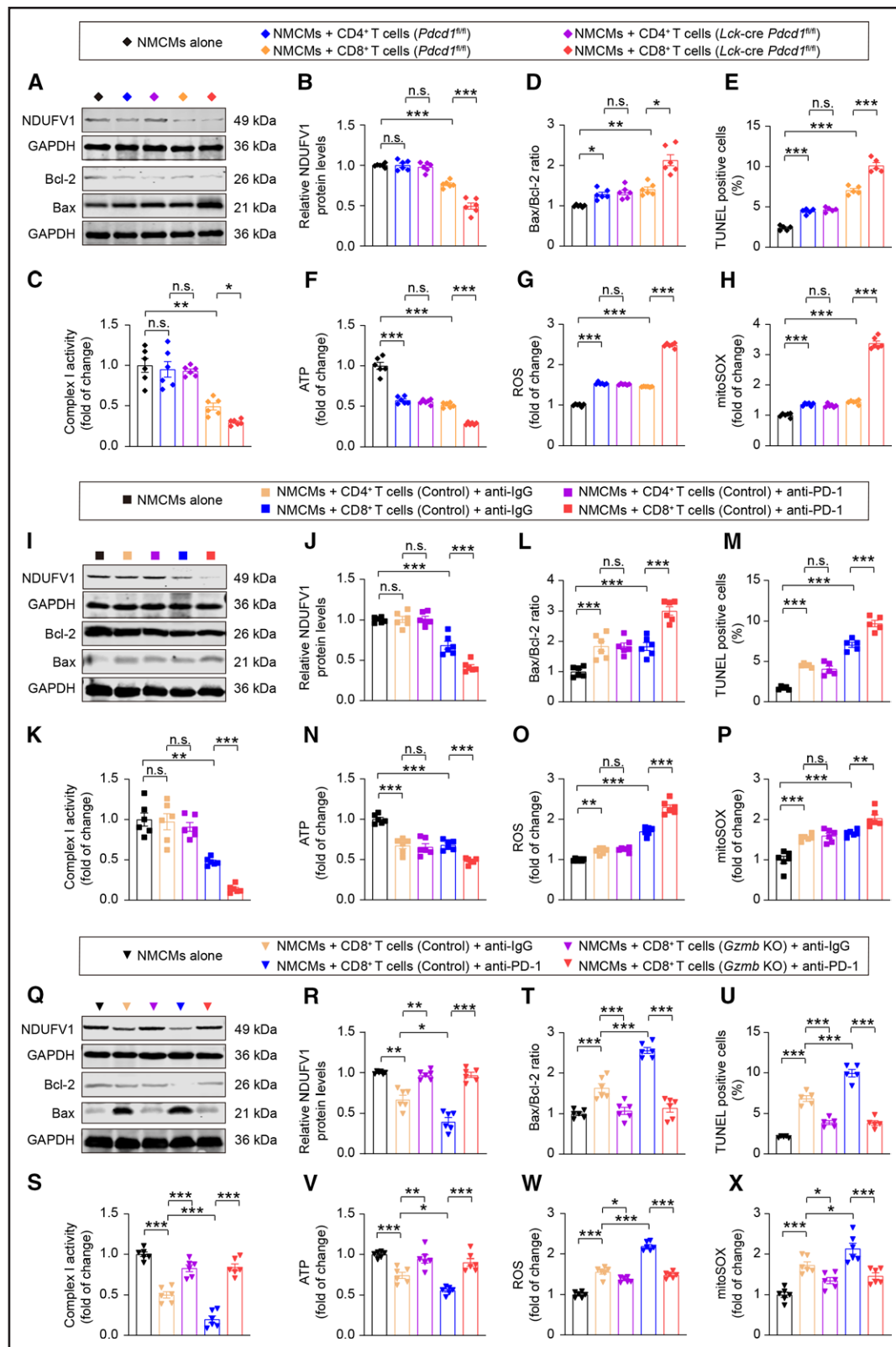


Figure 8. GZMB⁺ CD8⁺ T cells account for the impairment of mitochondrial respiratory chain complex I, ROS production, and apoptosis of cardiomyocytes.

A through **X**, Representative Western blots and quantification of NADH dehydrogenase (ubiquinone) flavoprotein 1 (NDUFV1), Bcl-2-associated X (Bax), and B-cell lymphoma-2 (Bcl-2) expression in neonatal mouse cardiomyocytes (NMCMs). The cells were cocultured with CD4⁺ or CD8⁺ T cells isolated from the indicated experimental groups (**A**, **B**, **D**, **I**, **J**, **L**, **Q**, **R**, and **T**; n=6 per group). Quantitative analysis of mitochondrial complex I activity in NMCMs (**C**, **K**, and **S**; n=6 per group); Quantification of TUNEL (TdT-mediated dUTP nick-end labeling)-positive (*Continued*)

Figure 8 Continued. NMCMs cocultured with CD4⁺ or CD8⁺ T cells isolated from the indicated experimental groups. (**E, M, and U**; n=6 per group). Quantitative analysis of the ATP levels in NMCMs (**F, N, and V**; n=6 per group). Quantitative analysis of the reactive oxygen species (ROS) levels after CM-H2DCFDA staining and mitochondrial ROS levels after MitoSOX staining (**G, H, O, P, W, and X**; n=6 per group). Data are presented as mean±SEM. Statistics: 1-way ANOVA followed by the Tukey post hoc multiple-comparisons tests (**B, E, G, H, J, L** through **N, P, S** through **W**, and **X**); Brown-Forsythe ANOVA followed by the Dunnett T3 multiple-comparisons test (**C, D, F, K, O**, and **R**). GZMB indicates granzyme B; and n.s., no significance between the 2 indicated groups. **P*<0.05. ***P*<0.01. ****P*<0.001.

myocarditis.³⁵ We demonstrated that PD-1 blockade or deletion resulted in an increased myocardial infiltration of both CD4⁺ T cells and CD8⁺ T cells in TAC mice. We suggest that the CXCL9/CXCL10-CXCR3 chemotaxis-mediated increase in myocardial infiltration of GZMB⁺ CD8⁺ T cells, but not CD4⁺ T cells, was the determinant for worsening TAC-induced cardiac manifestations due to the following reasons. High concentrations of CXCL9/CXCL10, the T cell-chemotactic chemokines, strongly correlate with ICI-associated immune-related adverse events in humans.³⁶ CXCR3, the receptor for CXCL9 and CXCL10, is expressed predominantly on activated CD8⁺ T cells and is crucial for CD8⁺ T-cell trafficking to the sites of inflammation.^{37,38} Consistent with the reanalysis data generated from patients with hypertrophic cardiomyopathy and DCM (GSE141910),^{18,19} we showed that TAC increased the expression of *Cxcl9/Cxcl10* and *Cxcr3* in mouse LVs, which was further elevated by PD-1 blockade or T cell- or CD8⁺ T cell-specific *Pdcd1* deletion. Moreover, *Cxcl9* mRNA expression in the LVs of TAC mice was significantly decreased by *Cxcr3* KO.³⁹ Of note, the increased expression levels of *Cxcl9/Cxcl10* were in parallel with PD-1 blockade or T cell- or CD8⁺ T cell-specific *Pdcd1* deletion-induced myocardial infiltration of CXCR3⁺ CD8⁺ T cells. The anti-PD-1 antibody blockade-induced increases in myocardial infiltration of CD8⁺ T cells and *Cxcl9*, *Cxcl10*, *Cxcr3*, *Cd3e*, *Gzmb*, and *Prf1* levels in TAC mice were diminished by AMG487 or *Cxcr3* KO. However, our data favor the notion that the AMG487- or *Cxcr3* KO-induced reduction of *Gzmb* and *Prf1* mRNA levels in the LVs of TAC mice is due primarily to blocking chemotaxis of CD8⁺ T cells rather than a direct modulation of its cytotoxic function. Moreover, Ang II-induced increase in CFs-derived *Cxcl9/Cxcl10* expression was linked to AT1 receptor/nuclear factor- κ B axis, which may act as an initial event to enhance myocardial infiltration of CXCR3⁺ CD8⁺ T cells under TAC because pressure overload elevates cardiac local Ang II levels.^{40,41}

Our data obtained from patients with anti-PD-1 antibody therapy-associated possible myocarditis indicate that the serum expression levels of GZMB/perforin and GZMB/perforin in peripheral CD8⁺ T cells were significantly increased. However, these data have limitations in that the experimental models are distinct from the clinical cases, albeit a few cases have hypertension history and thickened septum. The results from the US FDA Adverse Event Reporting System database (a large-scale pharmacovigilance resource) are supportive of the notion that the use of anti-PD-1 antibodies significantly elevates

the risk of HF-related adverse events in patients with preexisting hypertension and diabetes. These data suggest that ICI treatment of patients with cancer with preexisting cardiovascular conditions may need proactive surveillance, although the data obtained from the US FDA Adverse Event Reporting System database present neither severity-stratified and longitudinal analyses nor a direct correlation between circulating immune signatures and myocardial pathology. We speculate that the aforementioned data may provide indirect evidence to support our hypothesis because anti-PD-1 antibody therapy-induced cardiotoxicities occurred concomitantly with the increased peripheral GZMB and perforin levels. Nevertheless, we show that the CXCL9/CXCL10-CXCR3 axis acts as a critical driving force for recruiting cytotoxic CD8⁺ T cells into TAC mouse hearts. Because AMG487 with clinically proven safety^{42,43} has shown a significant inhibitory effect on breast cancer and colon cancer progression,^{42,43} it may serve as a potential agent for preventing ICI-associated pathological cardiac remodeling and HF.

Mechanistically, PD-1 blockade- or deletion-induced impairment of cellular respiration and oxidative phosphorylation pathways, known to affect the progression of pathological HF,⁴⁴ plays a critical role in exacerbating TAC-induced cardiac injury. GZMB, as a protease, could directly cleave NDUFV1 and impair mitochondrial complex I activity, subsequently elevating mitochondrial ROS and causing apoptosis in human leukemia K562 cells.²⁸ We show that among the tested mitochondrial complex I subunits, GZMB selectively suppresses NDUFV1 protein levels. However, GZMB may also cleave or functionally impair other mitochondrial targets without altering their steady-state protein abundance. Of note, incubation of NMCMs with both recombinant perforin and GZMB markedly reduced the enzymatic activities of mitochondrial complexes I, III, and V but not complexes II or IV, suggesting that GZMB indeed impairs mitochondrial respiratory function of cardiomyocytes by targeting both steady-state NDUFV1 protein levels and multiple complexes. It is important to note that PD-1 blockade or T cell-specific *Pdcd1* deletion specifically promotes CD8⁺ T cell-mediated production and release of GZMB, although the myocardial infiltration of both CD4⁺ T cells and CD8⁺ T cells was significantly increased in LVs of TAC mice. This is not surprising because CD4⁺ T cells promote the transition from compensated hypertrophy to HF after TAC surgery.³³ No further adverse effects were

observed when NMCMs were cocultured with CD4⁺ T cells in the presence of anti-PD-1 antibody or CD4⁺ T cells isolated from T cell-specific *Pdcd1* KO mice. In contrast, coculture of NMCMs with CD8⁺ T cells, isolated from littermate control mice, significantly inhibited NDUFV1 expression, impaired complex I activity, and led to apoptosis. The aforementioned effects were further augmented by either PD-1 blockade or T cell-specific *Pdcd1* KO. These data are comparable with recent studies showing that selective depletion of CD8⁺ T cells could prevent ICI-related cardiotoxicity in anti-PD-1 antibody-treated tumor-bearing mice¹³ and that cardiac-specific overexpression of NDUFS1, a subunit of mitochondrial complex I, decreases myocardial infarction-induced oxidative stress and apoptosis.⁴⁵

Mitochondrial complex I is an important site of ROS production,⁴⁶ and impaired complex I activity leads to increased mitochondrial ROS generation in ischemia/reperfusion mouse hearts.⁴⁷ Complex I deficiency-associated impairment of mitochondrial respiration causes Bax-dependent neuronal apoptosis through mitochondrial oxidative damage.⁴⁸ GZMB has been identified as a major cytotoxic protein that cleaves BH3-interacting domain death agonist into GZMB-truncated BH3-interacting domain death agonist, leading to the activation of Bax/Bak.⁴⁹ Thus, we suggest that the increased GZMB/perforin by the anti-PD-1 antibody-mediated activation of cytotoxic CD8⁺ T cells may inhibit NDUFV1 expression and enzyme activities of mitochondrial respiratory chain complexes I, III, and V to cause cardiomyocyte apoptosis, thereby leading to PD-1 suppression-induced worsening of cardiac injury in TAC mice. Notably, although adeno-associated virus serotype 9-mediated cardiac-specific GZMB overexpression exacerbates TAC-induced HF, GZMB neutralization rescued the worsening effects of T cell-specific *Pdcd1* KO on TAC-induced pathological cardiac remodeling and HF. Similar to what was observed in an acute myocardial infarction model,⁵⁰ the increased local GZMB likely derived from the myocardial infiltrated cytotoxic CD8⁺ T cells leads to cardiomyocyte apoptosis and HF in a perforin-dependent manner. Experiments conducted in CD8⁺ T cell-specific *Gzmb* KO mice are required to definitively address whether CD8⁺ T cell-derived GZMB is solely responsible for exacerbating TAC-induced cardiac remodeling and HF.

CONCLUSION

We identified that CXCL9/CXCL10-CXCR3 chemotaxis drives the myocardial infiltration of cytotoxic CD8⁺ T cells and that GZMB and perforin, likely derived from anti-PD-1 antibody-activated cytotoxic CD8⁺ T cells, are responsible for worsening cardiac injury under TAC. Targeting the CXCL9/CXCL10-CXCR3 axis or GZMB might be a novel strategy to prevent ICI-associated

cardiotoxicity in the population with preexisting cardiac pathological conditions.

ARTICLE INFORMATION

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Author Contributions

Z.-W.P., Y.L., and Z.-R.Z. conceived and designed the study. M.-M.W., Y.S., Y.Z., Y.-C.Y., Z.-X.H., S.Z., X.W., H.X., M.Y., C.M., B.-W.Z., X.-L.L., J.-Y.C., D.Z., Z.H., M.Z., H.-T.X., X.B., S.-Q.H., J.-J.M., J.S., C.L., Q.-S.W., Q.S., J.Z., and C.-J.Y. conducted experiments and acquired data. M.-M.W., Y.S., and Y.Z. analyzed data. J.-Y.L. and X.-H.H. collected and analyzed FDA Adverse Event Reporting System data. H.-X.L., T.L., L.-Y.Y., H.-B.J., and Y.J. provided critical insights to the study and revised the manuscript. Z.-W.P., Y.L., and Z.-R.Z. revised the manuscript and supervised the study. All authors read and approved the final version of the manuscript.

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Disclosures

None.

Supplemental Material

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